Title: Data Cleanser
Duration: 6 Hours

🎯 Objective

The aim of this project is to practice **Data Preprocessing and Feature Engineering** with a strong emphasis on **handling missing values** and **outlier detection/removal**. Students will apply different **imputation strategies** and **outlier treatment techniques** to clean a dataset and prepare it for machine learning tasks.

📑 Problem Statement

You are working as a **Data Analyst** for a healthcare company. You receive a dataset containing **patient health records** with missing values and outliers due to inconsistent reporting and measurement errors. Your task is to **clean the dataset** using various **missing data imputation methods** and **outlier handling techniques** so that it becomes suitable for downstream machine learning (predicting risk of heart disease).

The dataset contains:

- **Patient demographic details** (Age, Gender, Region).
- **Medical attributes** (BMI, Blood Pressure, Cholesterol Level, Glucose Level).
- **Target variable** (Disease risk: 0 = Low, 1 = High).

📁 Dataset Structure: Patient Health Records (Generate from an AI tool)

Field Name	Data Type	Description	Possible Missingness/Outliers

patient_id	String/Int	Unique identifier for each patient	No missing, no outliers
age	Integer	Age of the patient (in years)	Missing values in some records
gender	Categorical	Gender of patient: Male / Female	Missing values → test category imputation
region	Categorical	Region of residence: North / South / East / West	Missing values → test most frequent imputation
bmi	Float	Body Mass Index (weight/height^2)	Missing values + outliers (very high/low BMI)
blood_pressure	Float	Average systolic blood pressure (mmHg)	Outliers (extreme high values)
cholesterol	Float	Cholesterol level (mg/dL)	Missing values + outliers (extreme low/high)
glucose	Float	Fasting glucose level (mg/dL)	Missing values + outliers (very high spikes)
disease_risk	Binary Int	Target variable: 0 = Low Risk, 1 = High Risk	No missing, used as target

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🧩 How Students Will Use It

- **Missing Values Section:**
  - `age`, `gender`, `region`, `bmi`, `cholesterol`, `glucose` → intentionally left with some missing entries to test:
    - Simple Imputer (numerical + categorical)
    - Most Frequent Imputation
    - Missing Indicator + Random Sample Imputation
    - KNN Imputer
    - MICE Algorithm

- **Outliers Section:**

- `bmi`, `blood_pressure`, `cholesterol`, `glucose` → contain synthetic outliers (extreme high/low values) to test:
  - Z-score method
  - IQR method
  - Percentile method
  - Winsorization technique

- **Target Variable:**
  - `disease_risk` is kept intact (0/1) for downstream **machine learning framing**.

✅ With this structure, students should:

- Run **missing data analysis** → impute with various strategies.
- Run **outlier detection** → apply all 4 methods and compare results.
- Produce a **clean dataset + report** suitable for machine learning.

📝 Tasks

Part A: Handling Missing Values

1. Identify missing values and provide a **summary report** (percentage per column).
2. Apply the following **imputation techniques** and compare results:
   - **Simple Imputer (Numerical)**: Replace missing BMI with mean or median.
   - **Simple Imputer (Categorical)**: Replace missing Region with the most frequent value.
   - **Most Frequent Imputation**: Replace missing Gender with the most common category.
   - **Missing Indicator + Random Sample Imputation**: Create binary indicator columns for missingness and use random sampling to impute values.
   - **KNN Imputer**: Apply k-Nearest Neighbors imputation for multivariate imputation.
   - **MICE Algorithm**: Perform chained equations imputation for multiple variables simultaneously.

Part B: Handling Outliers

3. Detect and remove outliers using:
    - **Z-score method**: Identify patients with extreme cholesterol or glucose values.
    - **IQR method**: Use interquartile range to detect unusual BMI values.
    - **Percentile method**: Cap values below 1st percentile and above 99th percentile.
4. Apply **Winsorization** to cap extreme outliers instead of removing them.
5. Compare dataset shape and summary **before vs after outlier treatment**.

Part C: Final Clean Dataset

6. Present the final **cleaned dataset** with:
    - All missing values treated appropriately.
    - All outliers handled using suitable methods.

7. Provide a **brief report** explaining:
    - Which imputation strategy was most effective.
    - Which outlier handling method preserved data quality best.
    - How data cleaning improved dataset usability.

📌 Expected Outcome

By the end of this project, students will:

- Understand and apply **multiple missing value imputation strategies**.
- Implement **KNN and MICE** for multivariate imputations.
- Detect and treat outliers using **Z-score, IQR, Percentiles, and Winsorization**.
- Learn to compare **before vs after** dataset quality.
- Deliver a **machine learning–ready dataset** with a profiling summary.

✅ Submission Guidelines

- Include practical implementation in a Jupyter Notebook and screenshots.
- Label all charts (if any) clearly and write short interpretations under each result.
- **GitHub Repository:**
    - Create a GitHub repository to host your project.
    - Upload your project files, including source code, and documentation to the repository.

- Add a document (PDF) explaining theory concepts with definitions (if any).
- Ensure that you provide a clear and descriptive README.md file.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!

Data Cleanser

Data Preprocessing and Feature Engineering

BRING ON YOUR CODING ATTITUDE